INCH-POUND

FF-B-2844/04
15 FEBRUARY 1994

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, SINGLE ROW, RADIAL, NON-FILLING SLOT, MINIATURE SIZE, INSTRUMENT TYPE (CORROSION RESISTANT STEEL)

The General Services Administration has authorized the use of this specification by all Federal Agencies.

The complete requirements for procuring the bearings described herein shall consist of this document and the latest described FF-B-2844/GEN.

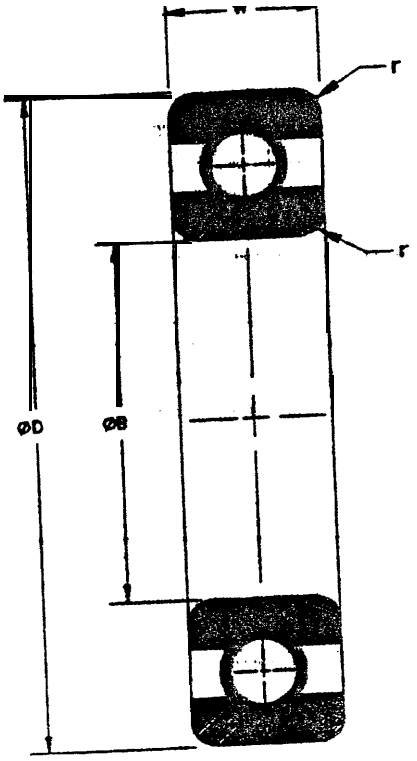


FIGURE 1.
OPEN(NO SHIELDS OR SEALS)

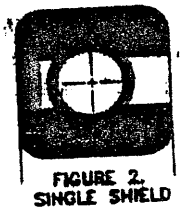


FIGURE 2.
SINGLE SHIELD

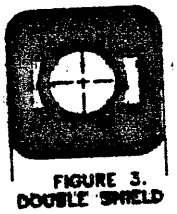


FIGURE 3.
DOUBLE SHIELD

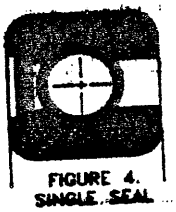


FIGURE 4.
SINGLE SEAL

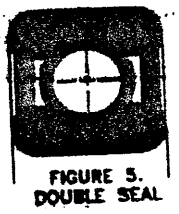


FIGURE 5.
DOUBLE SEAL

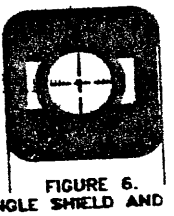


FIGURE 6.
SINGLE SHIELD AND SEAL

BEARING CONFIGURATIONS

FSC 3110

AMSC/NA

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REQUIREMENTS

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the bore code in Table I.

2. MATERIAL

2.1 Ring and ball material. The rings and balls of bearings covered by this specification sheet shall be made of corrosion resistant steel, 440C (UNS S44004), conforming to QQ-S-763.

2.1.1 Passivation. Passivation shall be accomplished in accordance with ASTM A380 on all bearing components fabricated from corrosion resistant steel after completion of removing operations and prior to assembly.

2.2 CLOSURES.

2.2.1 Shields. The shields shall be fabricated from corrosion resistant steel conforming to ASTM A580 Condition A, ASTM A313 Condition C or QQ-S-763 and shall be removable or non removable. The number of shields required shall be specified in the part number by using the code listed in Table IV.

2.2.2 Seals. The seals shall be fabricated from any material unless otherwise specified and shall be impervious contact type (removable non-removable). The number of seals required shall be specified in the part number by using the codes listed in Table IV.

2.3 Cage material. Cages shall be manufactured from any material unless otherwise specified. When manufactures standard is specified, corrosion resisting steel shall be used. Crown retainers shall be AISI 410 and ribbon retainers shall be either AISI 430, 305 or 302 in accordance with QQ-S-766. The type of cage required shall be specified in the part number by using the codes listed in Table II.

3. METROLOGY

3.1 Radial internal clearance. The tolerance limits for radial internal clearance are as follows and shall be specified in the part number by using the codes listed in Table III.

Clearance in 0.0001 inches	
Tight clearance	1 - 3
Normal clearance	2 - 5
Loose clearance	5 - 8
Extra loose clearance	8 - 11

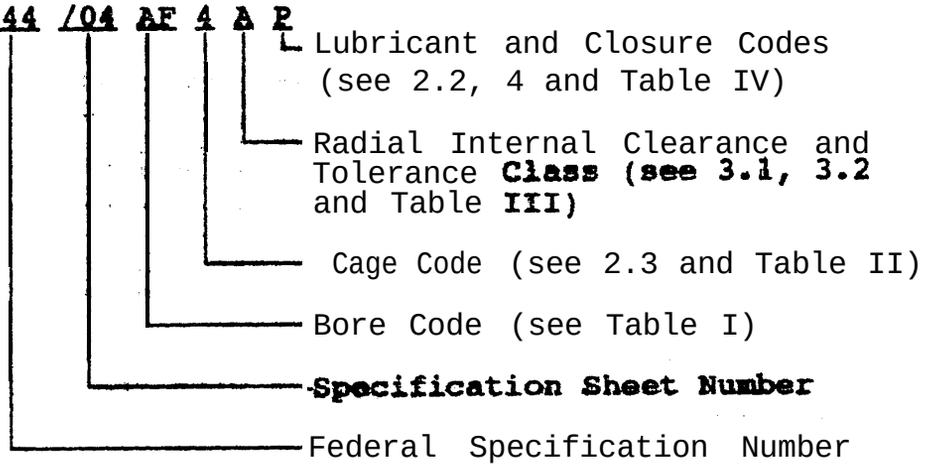
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3.2 **Tolerance class.** The tolerance class limits shall be the followings ABBC-1 In Accordance With (IAW) ANSI/AFBMA Std-20; ABEC 3P IAW ANSI/AFBMA STD 12.2. The tolerance class shall be specified in the part number by using the codes listed in Table III.

4. **Lubricant or preservative compound.** The lubricant or preservative Compound shall be specified in the part number by using the codes listed in Table IV. Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings, the bearing part number shall include the “Cm, “F” or "L” suffix from Table IV. For double closure or single closure bearings that require lubricant, the bearing part number shall include the preferred suffix from Table IV es required. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 40 percent filled.

5. Part number. The part number consists of the Federal specification prefix letters FFB, the basic number of the Federal Specification, the specification sheet number, bore code, and alpha or numeric codes for cage, radial internal clearance and tolerance class, lubricant and closures.

EXAMPLE: **FFB2844 /04 AF 4 A P**



FFB2844/04AF4AP indicates - Bearing Bore 0.0781 inches; outside diameter 0.2500 inches; width 0.1406 inches; cage type, machined non metallic; radial internal clearance, tight; tolerance class, ABEC 1; single shield and single seal; lubricant in accordance with DoD-G-24508.

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TABLE I. BOUNDARY DIMENSIONS 1/

Bore Code	OB Bore Diameter (inch)	OB Bore Diameter (MM)	OD Outside Diameter	W Width	Chamfer Min. 2/	Dia. Series 3/	Dynamic load Rating (lbs)
AA	0.0469	1.1906	0.1562	0.0625	.003	9	18
AB	0.0469	1.1906	0.1562	0.0937	.003	9	18
AC	0.0550	1.3970	0.1875	0.0781	.003	9	28
AD	0.0550	1.3970	0.1875	0.1094	.003	9	28
AE	0.0781	1.9844	0.2500	0.0937	.003	9	36
AF	0.0781	1.9844	0.2500	0.1406	.003	9	36
AG	0.0937	2.3812	0.1875	0.0625	.003	7	24
AH	0.0937	2.3812	0.1875	0.0937	.003	7	24
AJ	0.0937	2.3812	0.2500	0.0937	.003	9	36
AK	0.0937	2.3812	0.2500	0.1094	.003	9	36
AL	0.0937	2.3812	0.3125	0.1094	.003	0	54
AM	0.0937	2.3812	0.3125	0.1406	.003	0	54
AN	0.1250	3.1750	0.2500	0.0937	.003	7	36
AP	0.1250	3.1750	0.2500	0.1094	.003	7	36
AQ	0.1250	3.1750	0.3125	0.1094	.003	9	54
AR	0.1250	3.1750	0.3125	0.1406	.003	9	54
AT	0.1250	3.1750	0.3750	0.1094	.005	2	76
AU	0.1250	3.1750	0.3750	0.1406	.005	2	76
AW	0.1562	3.9687	0.3125	0.1094	.003	7	37
AX	0.1562	3.9687	0.3125	0.1250	.003	7	37
AY	0.1875	4.7625	0.3125	0.1094	.003	7	43
BA	0.1875	4.7625	0.3125	0.1250	.003	7	43
BB	0.2500	6.3500	0.3750	0.1250	.003	7	36
BC	0.3125	7.9375	0.5000	0.1562	.003	7	99

1/ All dimensions are in inches unless otherwise specified.
2/ The Chamfer on bearings will clear a maximum fillet radius equal to a minimum chamfer.
3/ Diameter Series is a soft metric conversion of the inch outside diameter and width in comparsion with the boundary dimension in AFBMA Std 20.

TABLE II. CAGES

CODE	CAGE
1	Manufacturers Standard 1/
2	Molded Plastic
3	Machined Brass or Bronze
4	Machined Non Metallic (phenolic)

1/ One piece crown or two piece ribbon

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TABLE III. RADIAL INTERNAL CLEARANCE AND TOLERANCE CLASS

Code	Internal Clearance	Tolerance
A B C D	Tight Normal Loose Extra Loose	ABEC 1
E F G H	Tight Normal Loose Extra Loose	ABEC 3P

TABLE IV. **LUBRICANTS AND CLOSURES**

code	Lubricant	Closure
A B C	DOD-G-24508 MIL-G-81322 Preservative 1/	OPEN (FIG 1)
D E F	DOD-G-24508 MIL-G-81322 Preservative 1/	SINGLE SHIELD (FIG 2)
G H	DOD-G-24508 MIL-G-81322	DOUBLE SHIELD (FIG 3)
J K L	DOD-G-24508 MIL-G-81322 Preservative 1/	SINGLE SEAL (FIG 4)
M N	DOD-G-24508 MIL-G-81322	DOUBLE SEAL (FIG 5)
P R	DOD-G-24508 MIL-G-81322	SINGLE SHIELD AND SEAL (FIG 6)

1/ Preservative shall be in accordance with MIL-P-197.

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